

A GENERAL FRAMEWORK FOR HYPERGRAPH COLORING*

IAN M. WANLESS[†] AND DAVID R. WOOD[†]

Abstract. The Lovász Local Lemma is a powerful probabilistic technique for proving the existence of combinatorial objects. It is especially useful for coloring graphs and hypergraphs with bounded maximum degree. This paper presents a general theorem for coloring hypergraphs that in many instances matches or slightly improves upon the bounds obtained using the Lovász Local Lemma. Moreover, the theorem directly shows that there are exponentially many colorings. The elementary and self-contained proof is inspired by a recent result for nonrepetitive colorings by Rosenfeld [*Electron. J. Combin.*, 27 (2020), P3.43]. We apply our general theorem in the settings of proper hypergraph coloring, proper graph coloring, independent transversals, star coloring, nonrepetitive coloring, frugal coloring, Ramsey number lower bounds, and k -SAT.

Key words. hypergraph, coloring, local lemma

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1. Hypergraph coloring. In their seminal 1975 paper, Erdős and Lovász [26] introduced what is now called the Lovász Local Lemma. This tool is one of the most powerful probabilistic techniques for proving the existence of combinatorial objects. The authors' motivation was hypergraph coloring. A *hypergraph* G consists of a set $V(G)$ of *vertices* and a set $E(G)$ of *edges*, each of which is a subset of $V(G)$. A *coloring* of a hypergraph G is a function that assigns a “color” to each vertex of G . A coloring of G is *proper* if no edge of G is monochromatic. The *chromatic number* $\chi(G)$ is the minimum number of colors in a proper coloring of G . The *degree* of a vertex v in a hypergraph G is the number of edges that contain v . A hypergraph is *r -uniform* if each edge has size r . Erdős and Lovász [26] proved (using the Lovász Local Lemma) that $\chi(G) \leq \lceil (4r\Delta)^{1/(r-1)} \rceil$ for every r -uniform hypergraph G with maximum degree Δ . The following result is a consequence of the strengthened Lovász Local Lemma first stated by Spencer [64]; see the book by Molloy and Reed [49] for a comprehensive treatment.

THEOREM 1 (see [26, 64]). *For every r -uniform hypergraph G with maximum degree Δ ,*

$$\chi(G) \leq \lceil (e(r(\Delta - 1) + 1))^{1/(r-1)} \rceil.$$

This paper presents a general theorem for coloring hypergraphs, which in the special case of proper hypergraph coloring (slightly) improves the upper bound in Theorem 1. Moreover, the proof directly shows that there are exponentially many such colorings. The proof uses a simple counting argument inspired by a recent result for nonrepetitive colorings by Rosenfeld [62], which in turn is inspired by the power series method for pattern avoidance [14, 52, 55].

It is well known that the proof of Theorem 1 works in the setting of list colorings, which we now introduce. Let G be a hypergraph. A *list-assignment* for G is a function

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[†]School of Mathematics, Monash University, Clayton, VIC 3800, Australia (ian.wanless@monash.edu, david.wood@monash.edu).

L that assigns each vertex v of G a set $L(v)$, whose elements are called *colors*. If $|L(v)| = c$ for each vertex v of G , then L is a c -list-assignment. An L -coloring of G is a function ϕ such that $\phi(v) \in L(v)$ for each vertex v of G . The *choosability* $\chi_{\text{ch}}(G)$ is the minimum integer c such that G has a proper L -coloring for every c -list-assignment L of G . For a list-assignment L of a hypergraph G , let $P(G, L)$ be the number of proper L -colorings of G .

The following theorem is our first contribution.

THEOREM 2. *For all integers $r \geq 3$ and $\Delta \geq 1$, and for every r -uniform hypergraph G with maximum degree Δ ,*

$$\chi_{\text{ch}}(G) \leq c := \left\lceil \left(\frac{r-1}{r-2} \right) ((r-2)\Delta)^{1/(r-1)} \right\rceil.$$

Moreover, for every c -list-assignment L of G ,

$$P(G, L) \geq ((r-2)\Delta)^{|V(G)|/(r-1)}.$$

We now compare the above-mentioned bounds. Since $(\frac{r-1}{r-2})^{r-2} < e$, it follows that $(\frac{r-1}{r-2})((r-2)\Delta)^{1/(r-1)} < (e(r-1)\Delta)^{1/(r-1)}$, and, assuming $\Delta \geq r-1$, the bound in Theorem 2 is slightly better than the bound in Theorem 1. The difference is most evident for small r . For example, if $r = 3$, then the bound in Theorem 2 is $\lceil 2\sqrt{\Delta} \rceil$ compared with $\lceil \sqrt{e(3\Delta-2)} \rceil$ from Theorem 1.

Several researchers have communicated to us that, with a little effort, one can conclude the existence of exponentially many colorings using the Lovász Local Lemma (or other methods), although as far as we are aware no general result of this nature has been published. One attraction of our proof is that it gives exponentially many colorings for free. Indeed, this stronger conclusion is key to enabling the simple proof. See [1, 2, 5, 8, 17, 30, 40, 51, 54] for more results on coloring hypergraphs with a given maximum degree or number of edges, and see [22, 23, 41, 45, 65, 66] for other theorems showing the existence of exponentially many colorings in various graph settings.

Theorem 2 is a special case of a more general result that we introduce in the following section. Then, in section 3, we apply this general result to a variety of coloring problems, including hypergraph coloring, graph coloring, independent transversals, star coloring, nonrepetitive coloring, frugal coloring, Ramsey number lower bounds, and k -SAT. Section 4 concludes by comparing our general result with other techniques, including the Lovász Local Lemma and entropy compression.

2. General framework. For a hypergraph G (allowing parallel edges), let $\mathcal{C}_G$ be the set of all colorings $\phi : V(G) \rightarrow \mathbb{Z}$. (For concreteness, we assume all colors are integers.) For an edge e of G , let $\mathcal{C}_e$ be the set of all colorings $\phi : e \rightarrow \mathbb{Z}$. An *instance* is a pair $(G, \mathcal{B})$ where G is a hypergraph and $\mathcal{B} = (\mathcal{B}_e \subseteq \mathcal{C}_e : e \in E(G))$. A coloring $\phi \in \mathcal{C}_G$ is $\mathcal{B}$ -bad if, for some edge $e \in E(G)$, we have that ϕ restricted to e is in $\mathcal{B}_e$. Every other coloring in $\mathcal{C}_G$ is $\mathcal{B}$ -good. For an integer $c \geq 1$, we say G is $(\mathcal{B}, c)$ -choosable if there is a $\mathcal{B}$ -good L -coloring of G for every c -list-assignment L of G . For a list-assignment L of G , let $P(G, \mathcal{B}, L)$ be the number of $\mathcal{B}$ -good L -colorings of G .

Fix an instance $(G, \mathcal{B})$ and consider an edge e of G . A subset $S \subseteq e$ determines $\mathcal{B}_e$ if any two colorings in $\mathcal{B}_e$ that agree on S are identical. For every vertex v in e , we assume that $\mathcal{B}_e$ is determined by some subset of $e \setminus \{v\}$. (Consider this assumption to be part of the definition of “instance.”) Then define the *weight* of (v, e) to be $|e| - 1 - |S|$, where S is a minimum-sized subset of $e \setminus \{v\}$ that determines $\mathcal{B}_e$. For each vertex v of G , let $E_k(v)$ be the number of pairs (v, e) with weight k .

For example, to model proper coloring in an r -uniform hypergraph G , for each edge e of G let $\mathcal{B}_e$ be the monochromatic colorings in $\mathcal{C}_e$. Then a coloring is $\mathcal{B}$ -good if and only if it is proper. For every edge e and every vertex v in e , if w is any vertex in $e \setminus \{v\}$, then $\{w\}$ determines $\mathcal{B}_e$, implying that (v, e) has weight $r - 2$.

THEOREM 3. *Let $(G, \mathcal{B})$ be an instance. Assume there exist a real number $\beta \geq 1$ and an integer $c \geq 1$ such that for every vertex v of G ,*

$$(1) \quad c \geq \beta + \sum_{k \geq 0} \beta^{-k} E_k(v).$$

Then G is $(\mathcal{B}, c)$ -choosable. Moreover, for every c -list-assignment L of G ,

$$P(G, \mathcal{B}, L) \geq \beta^{|V(G)|}.$$

Before proving Theorem 3 we make a couple of minor observations. If $\beta > 1$, then Theorem 3 guarantees exponentially many $\mathcal{B}$ -good colorings. If $\beta = 1$, then Theorem 3 guarantees at least one $\mathcal{B}$ -good coloring. In most applications, $\beta > 1$, but on one occasion the case $\beta = 1$ is of interest (see subsection 3.1). When applying Theorem 3 it is not necessary to determine the weight of a pair exactly; it suffices to determine a lower bound on the weight (because of the β^{-k} term in (1), where $\beta \geq 1$).

Theorem 3 is an immediate corollary of the following lemma. If $(G, \mathcal{B})$ is an instance with $\mathcal{B} = (\mathcal{B}_e : e \in E(G))$, and H is a subhypergraph of G , then $(H, \mathcal{B})$ refers to the instance $(H, (\mathcal{B}_e : e \in E(H)))$. Similarly, if L is a list-assignment for G , then we consider L (restricted to $V(H)$) to be a list-assignment for H .

LEMMA 4. *Let $(G, \mathcal{B})$ be an instance. Assume there exist a real number $\beta \geq 1$ and an integer $c \geq 1$ such that (1) holds for every vertex v of G . Then for every c -list-assignment L of G , for every induced subhypergraph H of G , and for every vertex v of H ,*

$$P(H, \mathcal{B}, L) \geq \beta P(H - v, \mathcal{B}, L).$$

Proof. We proceed by induction on $|V(H)|$. The base case with $|V(H)| = 1$ is trivial. Let H be an induced subhypergraph of G , and assume the claim holds for all induced subhypergraphs of G with less than $|V(H)|$ vertices. Let v be any vertex of H . Let X be the set of $\mathcal{B}$ -bad L -colorings of H that are $\mathcal{B}$ -good on $H - v$. Then

$$(2) \quad P(H, \mathcal{B}, L) = c P(H - v, \mathcal{B}, L) - |X|.$$

We now find an upper bound for $|X|$. For each L -coloring ϕ in X there is an edge $e \in E(H)$ containing v such that $\phi \in \mathcal{B}_e$ (if there are several options for e , fix a choice arbitrarily). Charge ϕ to (v, e) . Let X_k be the set of colorings in X that are charged to a pair with weight k . Consider ϕ in X_k charged to (v, e) . Let S be a minimum-sized subset of $e \setminus \{v\}$ that determines $\mathcal{B}_e$. Let $T := e \setminus S$. Then $|T| = k + 1$ and $v \in T$. Since ϕ is $\mathcal{B}$ -good on $H - v$, we know that ϕ is also $\mathcal{B}$ -good on $H - T$. Since S determines $\mathcal{B}_e$, the number of L -colorings in X_k charged to (v, e) is at most $P(H - T, \mathcal{B}, L)$. By induction,

$$P(H - v, \mathcal{B}, L) \geq \beta^k P(H - T, \mathcal{B}, L).$$

Thus the number of L -colorings in X_k charged to (v, e) is at most $\beta^{-k} P(H - v, \mathcal{B}, L)$. Hence, $|X_k| \leq E_k(v) \beta^{-k} P(H - v, \mathcal{B}, L)$ and

$$|X| = \sum_{k \geq 0} |X_k| \leq P(H - v, \mathcal{B}, L) \sum_{k \geq 0} E_k(v) \beta^{-k}.$$

By (2),

$$P(H, \mathcal{B}, L) \geq c P(H - v, \mathcal{B}, L) - P(H - v, \mathcal{B}, L) \sum_{k \geq 0} \beta^{-k} E_k(v).$$

By (1), $P(H, \mathcal{B}, L) \geq \beta P(H - v, \mathcal{B}, L)$, as desired. $\square$

3. Examples. In this section, we apply Theorem 3 for various types of (hyper)graph coloring problems and for k -SAT. In most cases, Theorem 3 matches or improves on the best known bound on the number of colors (as a function of maximum degree) and, in addition, shows that there are exponentially many colorings.

3.1. Proper coloring. First, we prove Theorem 2. Let G be an r -uniform hypergraph with maximum degree Δ where $r \geq 3$. For each edge e of G , let $\mathcal{B}_e$ be the monochromatic colorings in $\mathcal{C}_e$; then a coloring is $\mathcal{B}$ -good if and only if it is proper. Each pair (v, e) has weight $r - 2$, and $E_{r-2}(v) \leq \Delta$. Observe that (1) holds with $\beta := ((r - 2)\Delta)^{1/(r-1)}$ and $c := \lceil (\frac{r-1}{r-2})((r - 2)\Delta)^{1/(r-1)} \rceil$. Theorem 2 then follows from Theorem 3.

Now consider proper coloring in a graph with maximum degree Δ (the case $r = 2$ in the above). Then every pair (v, e) has weight 0, and $E_0(v) \leq \Delta$. Thus $c := \lceil \Delta + \beta \rceil$ satisfies (1). Theorem 3 with $\beta = 1$ says that every graph G with maximum degree Δ is $(\Delta + 1)$ -choosable. Theorem 3 with $\beta \geq 2$ says that for every $(\Delta + \beta)$ -list-assignment L of G there are at least $\beta^{|V(G)|}$ L -colorings. These well-known facts are easily proved by a greedy algorithm. It is interesting that the above general framework includes such statements (the Lovász Local Lemma does not). Note that the Local Action Lemma of Bernshteyn [9] is another general-purpose tool that implies $(\Delta + 1)$ -colorability; also see [10].

See [57, 58] for results about the number of 2-colorings in random hypergraphs and the number of k -colorings in random graphs.

3.2. Star coloring. A coloring ϕ of a graph G is a *star coloring* if it is proper and every bichromatic subgraph is a star forest; that is, there is no 2-colored P_4 (path on four vertices). The *star chromatic number* $\chi_{\text{st}}(G)$ is the minimum number of colors in a star coloring of G . Fertin, Raspaud, and Reed [28] proved (using the Lovász Local Lemma) that $\chi_{\text{st}}(G) \leq O(\Delta^{3/2})$ for every graph G , with maximum degree Δ , and that this bound is tight up to a $O(\log \Delta)$ factor. The best known bound is $\chi_{\text{st}}(G) \leq \sqrt{8}\Delta^{3/2} + \Delta$ proved by Esperet and Parreau [27] using entropy compression. Both these methods work for star choosability. We prove that the same bound holds with exponentially many colorings.

THEOREM 5. *Every graph G with maximum degree Δ is star $\lceil \Delta + \sqrt{8\Delta}(\Delta - 1) \rceil$ -choosable. Moreover, for every $\lceil \Delta + \sqrt{8\Delta}(\Delta - 1) \rceil$ -list-assignment L , there are at least $(\sqrt{2\Delta}(\Delta - 1))^{|V(G)|}$ star L -colorings of G .*

Proof. Define the following hypergraph G' with $V(G') = V(G)$: Introduce one edge $e = \{v, w\}$ to G' for each edge vw of G , where $\mathcal{B}_e$ is the set of L -colorings $\phi \in \mathcal{C}_e$ such that $\phi(v) = \phi(w)$, and introduce one edge $e = \{u, v, w, x\}$ to G' for each P_4 subgraph (u, v, w, x) of G , where $\mathcal{B}_e$ is the set of L -colorings $\phi \in \mathcal{C}_e$ such that $\phi(u) = \phi(w)$ and $\phi(v) = \phi(x)$. For any list-assignment L of G , note that G is star L -colorable if and only if $P(G', \mathcal{B}, L) \geq 1$. Also, the weight of each 2-element edge is 0, and the weight of each 4-element edge is 1. Thus $E_0(v) \leq \Delta$ and $E_1(v) \leq 2\Delta(\Delta - 1)^2$. Since (1) is satisfied with $\beta := \sqrt{2\Delta}(\Delta - 1)$ and $c := \lceil \Delta + \sqrt{8\Delta}(\Delta - 1) \rceil$, the result follows from Theorem 3. $\square$

3.3. Nonrepetitive graph coloring. Let ϕ be a coloring of a graph G . A path $(v_1, \dots, v_{2t})$ in G is *repetitively colored* by ϕ if $\phi(v_i) = \phi(v_{t+i})$ for each $i \in \{1, \dots, t\}$. A coloring ϕ of G is *nonrepetitive* if no path in G is repetitively colored by ϕ . The *nonrepetitive chromatic number* $\pi(G)$ is the minimum number of colors in a nonrepetitive coloring of G . The *nonrepetitive choice number* $\pi_{\text{ch}}(G)$ is the minimum integer c such that G has a nonrepetitive L -coloring for every c -list-assignment L of G . Alon et al. [4] proved that $\pi(G) \leq O(\Delta^2)$ for every graph with maximum degree Δ , and that this bound is tight up to a $O(\log \Delta)$ factor. The proof shows the same bound for π_{ch} . Several authors subsequently improved the constant in the $O(\Delta^2)$ term: to $36\Delta^2$ by Grytczuk [38], to $16\Delta^2$ by Grytczuk [37], to $(12.2 + o(1))\Delta^2$ by Harant and Jendrol' [39], and to $10.4\Delta^2$ by Kolipaka, Szegedy, and Xu [46]. All these proofs used the Lovász Local Lemma. Dujmović et al. [20] improved the constant to 1 by showing that for every graph G with maximum degree Δ ,

$$(3) \quad \pi(G) \leq \Delta^2 + O(\Delta^{5/3}).$$

The proof of Dujmović et al. [20] uses entropy compression; see [27, 34] for refinements and simplifications to the method. Equation (3) was subsequently proved using the Local Cut Lemma of Bernshteyn [10] and using cluster-expansion [7, 13]. Most recently, Rosenfeld [62] proved (3) with exponentially many colorings. His paper inspired the present work. We now show that the result of Rosenfeld follows from our general framework. Note that all of the above results hold in the setting of choosability.

THEOREM 6. *For every graph G with maximum degree Δ , if*

$$\beta := (1 + 2^{1/3}\Delta^{-1/3})(\Delta - 1)^2 \quad \text{and} \quad c := \lceil \beta + 2^{-2/3}\Delta^{5/3}(1 + 2^{1/3}\Delta^{-1/3})^2 \rceil,$$

then G is nonrepetitively c -choosable. Moreover, for every c -list-assignment L of G there are at least $\beta^{|V(G)|}$ nonrepetitive L -colorings of G .

Proof. Let G' be the hypergraph with $V(G') = V(G)$, where there is an edge $V(P)$ for each path P in G of even order. Here we consider a path to be a subgraph of G , so that a path and its reverse contribute one edge to G' . For each edge e of G' corresponding to a path P in G of order $2t$, let $\mathcal{B}_e$ be the set of L -colorings $\phi \in \mathcal{C}_e$ such that P is repetitively colored by ϕ . Thus G is nonrepetitively L -colorable if and only if $P(G', \mathcal{B}, L) \geq 1$.

Consider an edge e of G' corresponding to a path P in G on $2t$ vertices. For each vertex v in P , any coloring $\phi \in \mathcal{B}_e$ is uniquely determined by ϕ restricted to the t vertices in the half of P not containing v . Hence, (v, e) has weight $t - 1$. Every vertex of G is in at most $t\Delta(\Delta - 1)^{2t-2}$ paths on $2t$ vertices. So $E_{t-1}(v) \leq t\Delta(\Delta - 1)^{2t-2}$. Equation (1) requires

$$c \geq \beta + \sum_{t \geq 1} t\Delta(\Delta - 1)^{2t-2} \beta^{1-t}.$$

Define $\beta := (1 + \epsilon)(\Delta - 1)^2$ where $\epsilon > 0$ will be defined shortly. Equation (1) requires

$$c \geq (1 + \epsilon)(\Delta - 1)^2 + \Delta \sum_{t \geq 1} t(1 + \epsilon)^{-t+1} = (1 + \epsilon)(\Delta - 1)^2 + \epsilon^{-2}(1 + \epsilon)^2 \Delta.$$

Define $\epsilon := 2^{1/3}\Delta^{-1/3}$ (to approximately minimize $(1 + \epsilon)(\Delta - 1)^2 + \epsilon^{-2}(1 + \epsilon)^2 \Delta$). Then (1) holds with c defined above, and the result follows from Theorem 3. $\square$

3.4. Frugal coloring. For an integer $k \geq 1$, a coloring ϕ of a graph G is k -frugal if ϕ is proper and $|\{w \in N_G(v) : \phi(w) = i\}| \leq k$ for every vertex v of G and for every color i , where $N_G(v)$ is the set of neighbors of v in G . A 1-frugal coloring of G corresponds to a proper coloring of G^2 . Hind, Molloy, and Reed [43] proved that for each integer $k \geq 1$ and sufficiently large Δ , every graph with maximum degree Δ has a k -frugal coloring with $\max\{(k+1)\Delta, \frac{e^3}{k}\Delta^{1+1/k}\}$ colors. An example due to Alon shows that this upper bound is within a constant factor of optimal [43]. In particular, for all $\Delta \geq k \geq 1$, Alon constructed a graph with maximum degree at most Δ that has no k -frugal coloring with $\frac{1}{2k}\Delta^{1+1/k}$ colors. Here we improve the constant in the upper bound without assuming that Δ is sufficiently large and with exponentially many colorings.

THEOREM 7. *For all integers $\Delta > k \geq 2$, let*

$$\beta := \left((k-1)\Delta \binom{\Delta-1}{k} \right)^{1/k} \quad \text{and} \quad c := \Delta + \left\lceil \frac{k\beta}{k-1} \right\rceil.$$

Then every graph G with maximum degree Δ has a k -frugal c -coloring. Moreover, for every c -list-assignment L of G , the number of k -frugal L -colorings of G is at least $\beta^{|V(G)|}$.

Proof. Let G' be the hypergraph with $V(G') = V(G)$, where every edge of G is an edge of G' , and $\{w_1, \dots, w_{k+1}\}$ is an edge of G' for every vertex v of G and set $\{w_1, \dots, w_{k+1}\} \subseteq N_G(v)$. In the latter case, we say the edge is *centered* at v . For every edge $e = \{v, w\}$ of G' , let $\mathcal{B}_e$ be the set of L -colorings $\phi \in \mathcal{C}_e$ such that $\phi(v) = \phi(w)$. For every edge $e = \{w_1, \dots, w_{k+1}\}$ of G' , let $\mathcal{B}_e$ be the set of L -colorings $\phi \in \mathcal{C}_e$ such that $\phi(w_1) = \phi(w_2) = \dots = \phi(w_{k+1})$. Then a coloring of G is k -frugal if and only if it is $\mathcal{B}$ -good.

For each edge $e = \{v, w\}$ of G' , both (v, e) and (w, e) have weight 0. Consider an edge $e = \{w_1, \dots, w_{k+1}\}$ of G' centered at v . For each $i \in \{1, \dots, k+1\}$, the pair (w_i, e) has weight $k-1$ since every coloring $\phi \in \mathcal{B}_e$ is determined by $\{w_j\}$ for any $j \neq i$.

Consider a vertex v of G . Then $E_0(v) \leq \Delta$. Now consider a pair (v, e) with nonzero weight. Then (v, e) has weight $k-1$, and $e = \{w_1, \dots, w_k, v\}$ is centered at some vertex u for some vertices $w_1, \dots, w_k \in N_G(u) \setminus \{v\}$. There are at most Δ choices for u and at most $\binom{\Delta-1}{k}$ choices for $w_1, \dots, w_k$. Thus $E_{k-1}(v) \leq \Delta \binom{\Delta-1}{k}$. Hence,

$$\beta + \sum_{i \geq 0} E_i(v) \beta^{-i} \leq \beta + \Delta + \Delta \binom{\Delta-1}{k} \beta^{1-k} = \Delta + \frac{k\beta}{k-1} \leq c.$$

The result follows from Theorem 3. $\square$

Since $k(k-1)^{-1+1/k} \rightarrow 1$ and $\binom{\Delta-1}{k}^{1/k} \leq \frac{e}{k}(\Delta-1)$, Theorem 7 implies the following corollary.

COROLLARY 8. *As $\Delta > k \rightarrow \infty$, for every $\lceil (e + o(1))\Delta^{1+1/k}/k \rceil$ -list-assignment L of a graph G with maximum degree Δ , the number of k -frugal L -colorings of G is at least $\beta^{|V(G)|}$.*

Note that Alon's example in [43] shows that Corollary 8 is within a factor of $2e + o(1)$ of optimal.

3.5. Independent transversals and constrained colorings. Consider a hypergraph G . A set $X \subseteq V(G)$ is *independent* if no edge of G is a subset of X .

Consider a partition $V_1, \dots, V_n$ of $V(G)$. A *transversal* of $V_1, \dots, V_n$ is a set X such that $|X \cap V_i| = 1$ for each i . Let $\ell : V(G) \rightarrow \{1, \dots, n\}$ be the function where $\ell(v) := i$ for each vertex $v \in V_i$. For $S \subseteq V(G)$, let $\ell(S) := \{\ell(v) : v \in S\}$. An edge e of G is *stretched* by $V_1, \dots, V_n$ if $|\ell(e)| = |e|$. The following theorem provides a condition that guarantees an independent transversal.

THEOREM 9. *Fix integers $r \geq 2$ and $t \geq 1$. For an r -uniform hypergraph G , let $V_1, \dots, V_n$ be a partition of $V(G)$ such that $|V_i| \geq t$ and at most $r^{-r}(r-1)^{r-1}t^{r-1}|V_i|$ stretched edges in G intersect V_i for each $i \in \{1, \dots, n\}$. Then there exist at least $(\frac{r-1}{r}t)^n$ independent transversals of $V_1, \dots, V_n$.*

Proof. Non-stretched edges do not influence whether a transversal is independent, so we may assume that every edge is stretched. We may also assume that $|V_i| = t$, since if $|V_i| > t$ and v is a vertex in V_i with maximum degree, then by removing v and its incident edges we obtain another hypergraph satisfying the assumptions. Let X be the hypergraph with $V(X) := \{1, \dots, n\}$, where for each edge $\{v_1, \dots, v_r\}$ of G there is an edge $e = \{\ell(v_1), \dots, \ell(v_r)\}$ in X . By assumption, each vertex i of X has degree at most $r^{-r}(r-1)^{r-1}t^{r-1}|V_i| = r^{-r}(r-1)^{r-1}t^r$. Let L be the list-assignment of X with $L(i) := V_i$ for each $i \in \{1, \dots, n\}$. For each edge e of X corresponding to edge $\{v_1, \dots, v_r\}$ of G , let $\mathcal{B}_e$ be the set consisting of the L -coloring ϕ of e with $\phi(\ell(v_j)) = v_j$ for each $j \in \{1, \dots, r\}$. Thus $\mathcal{B}$ -good L -colorings of X correspond to independent transversals of $V_1, \dots, V_n$. Since $\mathcal{B}_e$ is determined by $\emptyset$, each pair (i, e) has weight $r-1$. Define $\beta := \frac{r-1}{r}t$. Then

$$|L(i)| = t = \beta + \frac{(r-1)^{r-1}t^r}{r^r\beta^{r-1}} \geq \beta + \frac{E_{r-1}(i)}{\beta^{r-1}}.$$

Thus (1) holds, and the result follows from Theorem 3. $\square$

Erdős, Gyárfás, and Łuczak [24] study independent transversals in a particular family of sparse hypergraphs. They define an $[n, k, r]$ -hypergraph to be an r -uniform hypergraph G whose vertex set $V(G)$ is partitioned into n sets $V_1, \dots, V_n$, each with k vertices, such that every edge is stretched by $V_1, \dots, V_n$, and for every r -element subset S of $\{1, 2, \dots, n\}$ there is exactly one edge $e \in E(G)$ such that $\ell(e) = S$. Erdős, Gyárfás, and Łuczak [24] defined $f_r(k)$ to be the maximum integer n such that every $[n, k, r]$ -hypergraph has an independent transversal. Using the Lovász Local Lemma, they proved that if

$$(4) \quad e \left(\binom{n}{r} - \binom{n-r}{r} \right) < k^r,$$

then $f_r(k) \geq n$. Observe that for every $[n, k, r]$ -hypergraph G with partition $V_1, \dots, V_n$, for each $i \in \{1, \dots, n\}$, exactly $\binom{n-1}{r-1}$ edges of G intersect V_i . Thus Theorem 9 implies that if

$$(5) \quad \binom{n-1}{r-1} \leq \frac{(r-1)^{r-1}k^r}{r^r},$$

then $f_r(k) \geq n$. We now compare these last two results. Consider r to be fixed. As k grows, the largest n satisfying (4) or (5) also grows, so we can think of n being large

relative to r . Then,

$$\begin{aligned}
 & \frac{(r-1)^{r-1} \left[\binom{n}{r} - \binom{n-r}{r} \right]}{r^r \binom{n-1}{r-1}} \\
 &= \left(\frac{r-1}{r} \right)^{r-1} \frac{n}{r^2} \left[1 - \frac{(n-r)!^2}{n!(n-2r)!} \right] \\
 &= \left(\frac{r-1}{r} \right)^{r-1} \frac{n}{r^2} \left[1 - \prod_{i=0}^{r-1} \frac{n-r-i}{n-i} \right] \\
 &\geq \left(\frac{r-1}{r} \right)^{r-1} \frac{n}{r^2} \left[1 - \left(\frac{n-r}{n} \right)^r \right] \\
 &= \left(\frac{r-1}{r} \right)^{r-1} \left[1 - \binom{r}{2} \frac{1}{n} + \frac{n}{r^2} \sum_{i=2}^{\lceil r/2 \rceil} \left(\binom{r}{2i-1} \left(\frac{r}{n} \right)^{2i-1} - \binom{r}{2i} \left(\frac{r}{n} \right)^{2i} \right) \right] \\
 &\geq \left(\frac{r-1}{r} \right)^{r-1} \left[1 - \frac{r^2}{2n} + \frac{n}{r^2} \sum_{i=2}^{\lceil r/2 \rceil} \binom{r}{2i-1} \left(\frac{r}{n} \right)^{2i-1} \left(1 - \frac{(r-2i+1)r}{2in} \right) \right] \\
 (6) \quad &\geq \left(\frac{r-1}{r} \right)^{r-1} \left[1 - \frac{r^2}{2n} \right]
 \end{aligned}$$

if $n \geq r^2/4$. Also $(1-1/r)^{r-1} > 1/e$. Hence, if n is sufficiently large relative to r , then (6) will exceed $1/e$, and (5) implies (4). In other words, our bound on $f_r(k)$ is better when k is sufficiently large relative to r . Yuster [67, 68] used a different argument to get a better bound in the case of graphs ($r=2$).

Theorem 9 in the case of graphs says the following.

COROLLARY 10. *Fix an integer $t \geq 1$. For a graph G , let $V_1, \dots, V_n$ be a partition of $V(G)$ such that $|V_i| \geq t$ and there are at most $\frac{t}{4}|V_i|$ edges in G with exactly one endpoint in V_i for each $i \in \{1, \dots, n\}$. Then there exist at least $(\frac{t}{2})^n$ independent transversals of $V_1, \dots, V_n$.*

Corollary 10 immediately implies the following result (since the average degree out of V_i is at most the maximum degree).

COROLLARY 11. *For a graph G with maximum degree at most Δ , if $V_1, \dots, V_n$ is a partition of $V(G)$ such that $|V_i| \geq 4\Delta$ for each $i \in \{1, \dots, n\}$, then there exist at least $(2\Delta)^n$ independent transversals of $V_1, \dots, V_n$.*

We now compare Corollaries 10 and 11 with the literature. Reed and Wood [61] proved the weakening of Corollary 10 with $\frac{t}{4}$ replaced by $\frac{t}{2e}$ and with $(\frac{t}{2})^n$ replaced by 1, and Dvořák et al. [21] noted that Corollary 10 holds with $(\frac{t}{2})^n$ replaced by 1 (using different terminology). Similarly, Alon [3] proved the weakening of Corollary 11 with 4Δ replaced by $2e\Delta$ and with $(2\Delta)^n$ replaced by 1. The proofs of Reed and Wood [61] and Alon [3] used the Lovász Local Lemma, while the proof of Dvořák et al. [21] used the Local Cut Lemma. Using a different method, Haxell [42] proved the strengthening of Corollary 11 with 4Δ replaced by 2Δ , but with $(2\Delta)^n$ replaced by 1. The bound here of 2Δ is best possible [16, 68]. It is open whether $\frac{t}{4}$ in Corollary 10 can be improved to $\frac{t}{2}$; see [44]. See [33, 47, 67] for more on independent transversals in graphs.

These results are related to the following “constrained coloring” conjecture of Reed [59].

CONJECTURE 12 (see [59]). *Let L be a $(k+1)$ -list-assignment of a graph G such that for each vertex v of G and color $c \in L(v)$, there are at most k neighbors $w \in N_G(v)$ with $c \in L(w)$. Then there exists a proper L -coloring of G .*

Haxell [42] observed the following connection between constrained colorings and independent transversals. Consider an $f(k)$ -list-assignment L of a graph G . Let H be the graph with $V(H) := \{(v, c) : v \in V(G), c \in L(v)\}$, where $(v, c)(w, c) \in E(H)$ for each edge $vw \in E(G)$ and color $c \in L(v) \cap L(w)$. Let $H_v := \{(v, c) : c \in L(v)\}$. Then $(H_v : v \in V(G))$ is a partition of H with each $|H_v| \geq f(k)$ such that proper L -colorings of G correspond to independent transversals of $(H_v : v \in V(G))$. Now if we assume that for each vertex v and color $c \in L(v)$ there are at most k neighbors $w \in N_G(v)$ with $c \in L(w)$, then H has maximum degree at most k . Hence, the above-mentioned result of Alon [3] proves Conjecture 12 with $k+1$ replaced by $2ek$ (also proved by Reed [59]), and the above-mentioned result of Haxell [42] proves Conjecture 12 with $k+1$ replaced by $2k$. Bohman and Holzman [15] disproved Conjecture 12. The best asymptotic result, due to Reed and Sudakov [60], says that for each $\epsilon > 0$ there exists k_0 such that Conjecture 12 holds with $k+1$ replaced by $(1+\epsilon)k$ for all $k \geq k_0$. None of these results conclude that there are exponentially many colorings. Corollary 10 and the above connection by Haxell [42] imply the following result.

COROLLARY 13. *Fix an integer $t \geq 2$. Let L be a t -list-assignment of a graph G such that for each vertex v of G ,*

$$4 \sum_{w \in N_G(v)} |L(v) \cap L(w)| \leq t^2.$$

Then there exist at least $(\frac{t}{2})^{|V(G)|}$ proper L -colorings of G .

Taking $t = 4k$ we obtain the following result in the direction of Conjecture 12.

COROLLARY 14. *Let L be a $4k$ -list-assignment of a graph G such that for each vertex v of G and color $c \in L(v)$, there are at most k neighbors $w \in N_G(v)$ such that $c \in L(w)$. Then there exist at least $(2k)^{|V(G)|}$ proper L -colorings of G .*

The following stronger result can also be proved using a variant of Theorem 3.

THEOREM 15. *Let L be a list-assignment of a graph G such that for every vertex v of G ,*

$$(7) \quad |L(v)| \geq 4 \sum_{w \in N_G(v)} \frac{|L(v) \cap L(w)|}{|L(w)|}.$$

Then there exist at least $\prod_{v \in V(G)} \frac{|L(v)|}{2}$ proper L -colorings.

Proof. We proceed by induction on $|V(H)|$ with the following hypothesis: for every induced subgraph H of G , and for every vertex v of H ,

$$P(H, L) \geq \frac{|L(v)|}{2} P(H - v, L).$$

(The proof is very similar to that of Lemma 4 except that β depends on v ; in particular, $\beta_v = \frac{|L(v)|}{2}$.) The base case with $|V(H)| = 1$ is trivial. Let H be an induced subgraph of G , and assume the claim holds for all induced subgraphs of G with less than $|V(H)|$ vertices. Let v be any vertex of H . Let X be the set of improper L -colorings of H that are proper on $H - v$. Then

$$(8) \quad P(H, L) = |L(v)| P(H - v, L) - |X|.$$

We now find an upper bound for $|X|$. For $w \in N_G(v)$, let X_w be the set of colorings ϕ in X such that $\phi(v) = \phi(w)$. Each L -coloring in X is in some X_w . Thus,

$$|X| \leq \sum_{w \in N_G(v)} |X_w| \leq \sum_{w \in N_G(v)} P(H - v - w, L) |L(v) \cap L(w)|.$$

By induction, $P(H - v, L) \geq \frac{|L(v)|}{2} P(H - v - w, L)$. Hence,

$$|X| \leq \sum_{w \in N_G(v)} \frac{2|L(v) \cap L(w)|}{|L(w)|} P(H - v, L).$$

By (8),

$$P(H, L) \geq |L(v)| P(H - v, L) - \sum_{w \in N_G(v)} \frac{2|L(v) \cap L(w)|}{|L(w)|} P(H - v, L).$$

By (7), $P(H, L) \geq \frac{|L(v)|}{2} P(H - v, L)$, as desired. $\square$

Note that Theorem 15 immediately implies Corollary 13, taking $|L(v)| = 2t$ for each v .

3.6. Ramsey numbers. For integers $k, c \geq 2$, let $R_c(k)$ be the minimum integer n such that every edge c -coloring of K_n contains a monochromatic K_k . Ramsey [56] and Erdős and Szekeres [25] independently proved that $R_c(k)$ exists. The best asymptotic lower bound on $R_2(k)$ is due to Spencer [63, 64], who proved that

$$(9) \quad R_2(k) \geq \left(\frac{\sqrt{2}}{e} - o(1) \right) k 2^{k/2}.$$

More precisely, Spencer [63, 64] proved that if

$$(10) \quad e \binom{k}{2} \left(\binom{n-2}{k-2} + 1 \right) < 2^{\binom{k}{2}-1},$$

then there exists an edge 2-coloring of K_n with no monochromatic K_k , implying $R_2(k) > n$. Theorem 3 leads to an analogous result with the same asymptotics but with slightly better lower order terms. For a graph G and integer $k \geq 2$, let $D_k(G)$ be the maximum, taken over all edges $vw \in E(G)$, of the number of k -cliques in G containing v and w .

THEOREM 16. *Fix integers $k \geq 3$ and $c \geq 2$. Let $m := \binom{k}{2} - 1$. Then for every graph G with*

$$(11) \quad D_k(G) \leq \frac{(m-1)^{m-1} c^m}{m^m},$$

there exists an edge c -coloring of G with no monochromatic K_k . In fact, there exists at least $(D_k(G)(m-1))^{|E(G)|/m}$ such colorings.

Proof. Let G' be the hypergraph with $V(G') := E(G)$, where $S \subseteq E(G)$ is an edge of G' whenever S is the edge-set of a K_k subgraph in G . For each edge vw of G , let $L(vw) := \{1, \dots, c\}$. For each edge S of G' , let $\mathcal{B}_S$ be the set of monochromatic L -colorings of S . Thus $\mathcal{B}$ -good L -colorings of G' correspond to edge c -colorings of G

with no monochromatic K_k . Each pair (v, e) has weight $m - 1$, and $E_{m-1}(v) \leq D_k(G)$. Thus (1) holds if

$$(12) \quad c \geq \beta + D_k(G) \beta^{1-m}.$$

To minimize the right-hand side of this expression, define $\beta := (D_k(G)(m-1))^{1/m}$. Then (11) implies (12), and so the result follows from Theorem 3. $\square$

Applying Theorem 16 to a complete graph gives the following corollary.

COROLLARY 17. *For every integer $k \geq 3$ and $c \geq 2$, if $m := \binom{k}{2} - 1$ and*

$$\frac{m^m}{(m-1)^{m-1}} \binom{n-2}{k-2} \leq c^m,$$

then there exists an edge c -coloring of K_n with no monochromatic K_k , and $R_c(k) > n$.

Since $\frac{m^m}{(m-1)^{m-1}} < em = e(\binom{k}{2} - 1)$, Corollary 17 is slightly stronger than (10). While this improvement only changes the implicit lower order term in (9), we consider it to be of interest since it suggests a new approach for proving lower bounds on $R_c(k)$.

3.7. k -SAT. The k -SAT problem takes as input a Boolean formula ψ in conjunctive normal form, where each clause has exactly k distinct literals, and asks whether there is a satisfying truth assignment for ψ . The Lovász Local Lemma proves that if each variable is in at most $\frac{2^k}{ke}$ clauses, then there exists a satisfying truth assignment; see [31] for a thorough discussion of this topic. The following result (slightly) improves upon this bound (since $(\frac{k-1}{k})^{k-1} > \frac{1}{e}$) and, moreover, guarantees exponentially many truth assignments.

THEOREM 18. *Let ψ be a Boolean formula in conjunctive normal form, with variables $v_1, \dots, v_n$ and clauses $c_1, \dots, c_m$, each with exactly k literals. Assume that each variable is in at most $\Delta := \frac{2^k}{k} (\frac{k-1}{k})^{k-1}$ clauses. Then there exists a satisfying truth assignment for ψ . In fact, there are at least $(2 - \frac{2}{k})^n$ such truth assignments.*

Proof. Let G be the hypergraph with $V(G) = \{v_1, \dots, v_n\}$ and $E(G) = \{e_1, \dots, e_m\}$, where edge e_i consists of those variables in clause c_i . So G is k -uniform. Let $L(v_i) = \{0, 1\}$ for each vertex v_i . Let $\mathcal{B}_{e_i}$ be the set of L -colorings of e_i such that c_i is not satisfied. Satisfying truth assignments for ψ correspond to $\mathcal{B}$ -good L -colorings of G . Each pair (v, e) has weight $k - 1$. Thus $E_{k-1}(v) \leq \Delta$ and $E_i(v) = 0$ for all $i \neq k - 1$. Then (1) holds with $\beta := 2 - \frac{2}{k}$ and $c := 2$. The result follows from Theorem 3. $\square$

Note that Gebauer, Szabó, and Tardos [32] proved that if each variable is in at most $(1 - o(1)) \frac{2^{k+1}}{ke}$ clauses, then there exists a satisfying truth assignment, and that this bound is best possible up to the $o(1)$ term; see Harris [40] for further improvements. These results improve upon the bound in Theorem 18 by a factor of 2. However, Theorem 18 may still be of interest since it gives exponentially many satisfying assignments and is an immediate corollary of our general framework.

See [18] for bounds on the number of satisfying truth assignments in random k -SAT formulas.

4. Reflection. We now reflect on Theorem 3, which provides a general framework for coloring hypergraphs of bounded degree.

First, we discuss minimizing the number of colors in Theorem 3. To do so, one needs to minimize the right-hand side of (1), which is a Laurent series $Q(\beta)$ with

nonnegative integer coefficients. We assume that at least one edge has positive weight, since otherwise $Q(\beta)$ is linear. We also assume that the coefficients in $Q(\beta)$ grow slowly enough that it and its first two derivatives converge for all $\beta > R$ for some real number R . For example, when the weight of edges is bounded (which is true in every example in this paper outside of subsection 3.3), we are optimizing a Laurent polynomial and may take $R = 0$. Now, $Q''(\beta) > 0$ for all $\beta > R$, so we expect a unique minimum for $Q(\beta)$ on the interval $[R, \infty)$, say at $\beta = \beta_0$. Since $Q'(1) \leq 0$ (or $R > 1$), we must have $\beta_0 \geq 1$. Even when using a value of $\beta \neq \beta_0$, one still obtains a nontrivial result from Theorem 3. In fact, choosing $\beta > \beta_0$ may be desirable if one wants to find conditions under which there are more colorings than are guaranteed by taking $\beta = \beta_0$.

Compared with the Lovász Local Lemma, Theorem 3 has the advantage of directly proving the existence of exponentially many colorings, and it often gives slightly better bounds. The proof of Theorem 3 is elementary, and, as discussed above, (1) is often easier to optimize than the General Lovász Local Lemma.

Theorem 3 should also be compared with entropy compression, which is a method that arose from the algorithmic proof of the Lovász Local Lemma due to Moser and Tardos [50]. See [12, 19, 20, 27, 35] for examples of the use of entropy compression in the context of graph coloring. We expect that the results in section 3 can be proved using entropy compression. For example, see [34, Theorem 12] for a generic graph coloring lemma in a similar spirit to our Theorem 3 that is proved using entropy compression. However, we consider the proof of Theorem 3 and the proofs of results that apply Theorem 3 to be simpler than their entropy compression counterparts, which require nontrivial analytic techniques from enumerative combinatorics. On the other hand, an advantage of entropy compression is that it provides an explicit algorithm for computing the desired coloring, often with polynomial expected time complexity.

It is also likely that our results in section 3 can be proved using the Local Cut Lemma [10] or via cluster-expansion [13]. The advantage of Theorem 3 is the simplicity and elementary nature of its proof. See [6, 29] for results connecting the Lovász Local Lemma, entropy compression, and cluster expansion.

Finally, we mention a technical advantage of the Lovász Local Lemma and of entropy compression. In the setting of hypergraph coloring, the Lovász Local Lemma and entropy compression need only bound the number of edges that intersect a given edge, whereas Theorem 3 requires a bound on the number of edges that contain a given vertex (because the proof is by induction on the number of vertices).

Note added in proof. Groenland et al. [36] recently showed that the constant 4 in Corollary 10 is best possible, which thus resolves the open problem stated in the paragraph after Corollary 11. In other recent developments, Bernshteyn et al. [11], Pirot and Hurley [53], and Martinsson [48] used Rosenfeld's method to prove the Johansson–Molloy bound on the chromatic number of triangle-free graphs.

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